The optimal exponent for 4D cylindrical cartoon approximation

A candidate full answer to the conjecture in arXiv:2110.03221

13 August 2026

Status: candidate full result, likely valid. In the standard polynomial-depth model used in the cited optimality literature, the squared L^2 approximation exponent 1 for the source's 4D cylindrical cartoon class is optimal. Thus the source's $N^{-1}(\log N)^2$ upper bound is sharp in its power of N.

1. Conjecture and precise conclusion

For $A > 0$, the source defines $\mathcal{E}(A) \subset L^2(\mathbb{R}^4)$ from functions

$$f(x,t) = h_0(x)g_0(t) + h_1(x)\mathbf{1}_B(x)g_1(t), \qquad x \in \mathbb{R}^3, t \in \mathbb{R},$$

where the spatial boundary ∂B is a uniformly C^2 two-manifold and all amplitudes have bounded C^2 norm. Its cylindrical shearlet theorem gives

$$\sup_{f \in \mathcal{E}(A)} \|f - f_N\|_2^2 \le CN^{-1}(\log N)^2. \tag{1}$$

The following remark conjectures that the exponent one is optimal.

Remark. The decay estimate above is the same as the one found for 3-dimensional shearlets [22] which is the optimal rate in the class of 3-dimensional cartoon-like functions [22, 23] and is faster than the optimal decay rate $O(N^{-2/3})$ valid for the class of 4-dimensional cartoon-like functions (cf. [23]). We conjecture that $O(N^{-1})$ is indeed the optimal decay rate in the class of 4-dimensional cylindrical cartoon-like images. In particular, the decay rate of 4-d cylindrical shearlets is significantly faster than conventional 4-d wavelets whose decay rate is $O(N^{-1/3})$.

3.3. Arguments and constructions

The general structure of the proof of Theorem 3.1 is similar to the structure of [22]. However, to deal with the geometry of 4d cylindrical shearlets, we need to introduce new technical constructions and modify some critical steps of the original arguments, especially in the proofs of Theorems 3.3 and 3.4 below.

To measure the sparsity of shearlet coefficients, we introduce the weak-ℓ^p quasi-norm $\|\cdot\|_{w\ell^p}$ which, for a sequence $s = (s_\mu)_{\mu \in M}$, is defined as

$$\|s\|_{w\ell^p} = \sup_{N>0} N^{1/p}|s_\mu|_{(N)}$$

where $|s_\mu|_{(N)}$ is the $N-th$ largest entry in the sequence s. In [24], this norm is shown to be equivalent to

Figure 1: The optimality conjecture on page 10 of arXiv:2110.03221.

The word "optimal" requires a feasibility convention. We use exactly the standard one in the 3D cartoon/shearlet literature cited by the source: normalized dictionaries with adaptive N-term selection subject to polynomial-depth search. Without an information budget of this kind, an arbitrarily fine dense dictionary destroys every universal lower bound [2].

Theorem 1 (optimality of the cylindrical exponent)**.** *Under polynomial-depth search, no dictionary and selection scheme can satisfy, for any $\varepsilon > 0$,*

$$\sup_{f\in\mathcal{E}(A)} \|f - A_N f\|_2^2 \le C_\varepsilon N^{-1-\varepsilon} \qquad (N \ge 1).$$

Consequently (1) *has the optimal power exponent, up to its logarithmic factor.*

2. A 3D hard class inside the 4D class

Let $\mathcal{C}_3(A_0)$ be the ordinary class of 3D C^2 cartoon functions

$$u(x) = a_0(x) + a_1(x)\mathbf{1}_B(x), \qquad x \in \mathbb{R}^3,$$

with uniformly controlled amplitudes and C^2 surface boundary. Its standard information-theoretic optimality theorem states that no polynomial-depth scheme has squared L^2 error exponent strictly larger than one [2,3].

Choose a nonzero $g \in C_c^2((-1,1))$ and normalize it in $L^2(\mathbb{R})$. Define

$$Ju(x,t) = u(x)g(t). \tag{2}$$

Then $J : L^2(\mathbb{R}^3) \to L^2(\mathbb{R}^4)$ is an isometry. After multiplying the whole class $\mathcal{C}_3(A_0)$ by one fixed positive constant, chosen only from the C^2 bounds of g and A, its image under J is contained in $\mathcal{E}(A)$. Such a fixed rescaling does not change any approximation exponent.

The adjoint of J is the time-profile projection

$$(J^*F)(x) = \int_{\mathbb{R}} F(x,t)\overline{g(t)}\, dt. \tag{3}$$

It satisfies $J^*J = I$ and $\|J^*\| = 1$.

Lemma 1 (projection of schemes)**.** *Every polynomial-depth N-term scheme on $L^2(\mathbb{R}^4)$ induces a polynomial-depth N-term scheme on $L^2(\mathbb{R}^3)$ whose error on u is at most the original error on Ju.*

Proof. Let an approximant selected from a normalized 4D dictionary be

$$A_N(Ju) = \sum_{n=1}^{N} c_n \phi_{i_n}.$$

Apply J^* and use $J^*J = I$:

$$\left\| u - \sum_{n=1}^{N} c_n J^*\phi_{i_n} \right\|_2 \le \|Ju - A_N(Ju)\|_2. \tag{4}$$

Each projected atom has norm at most one. Delete zero atoms; for every nonzero one, normalize $J^*\phi_i$ and absorb its norm into its coefficient. The indices and their ordering have not changed, so the same polynomial search bound applies. Equation (4) proves the claim. □

3. Proof and scope

Proof of Theorem 1. Suppose a 4D scheme achieved squared error $C_\varepsilon N^{-1-\varepsilon}$ uniformly on $\mathcal{E}(A)$. Restrict it to the fixed scaled copy $J\mathcal{C}_3(A_0) \subset \mathcal{E}(A)$ and apply Lemma 1. The result is a polynomial-depth 3D scheme with the same squared-error exponent on $\mathcal{C}_3(A_0)$. This contradicts the known 3D optimality theorem. Hence no exponent larger than one is possible. Together with (1), this proves optimality of the power exponent. $\square$

The result does not remove the logarithmic factor in (1); it settles exactly the power-law conjecture. A later paper [4] proves a related projection-based optimality statement for a different $L^2(\mathbb{R}^3)$ model with two spatial variables and time, whose optimal exponent is two. A bounded exact-title and conjecture search on 13 August 2026 found no explicit later answer for the present 3D-space-plus-time class.

Human-review focus. Confirm that the source intends the polynomial-depth benchmark of its cited 3D literature; the fixed tensor subclass lies in $\mathcal{E}(A)$ after rescaling; and normalizing projected atoms preserves the selection budget.

References

[1] T. A. Bubba, G. Easley, T. Heikkilä, D. Labate, and J. P. Rodriguez Ayllon, *Efficient representation of spatio-temporal data using cylindrical shearlets*, J. Comput. Appl. Math. 429 (2023), 115206.
[2] D. L. Donoho, *Sparse components of images and optimal atomic decompositions*, Constr. Approx. 17 (2001), 353–382.
[3] G. Kutyniok, J. Lemvig, and W.-Q. Lim, *Shearlets and optimally sparse approximations*, arXiv:1106.1325.
[4] T. A. Bubba, T. Heikkilä, D. Labate, and L. Ratti, *Regularization with optimal space-time priors*, arXiv:2405.06337.